

The Convolution Operation

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Leads to:** Padding and strides ([Note 1043](#)); Pooling ([Note 1044](#)); CNN architecture (LeNet-5) ([Note 1045](#)); Convolutional neural network (CNN) ([Note 1046](#)); Backpropagation in a CNN ([Note 1047](#)); Visualising what a CNN learns ([Note 1052](#)).

1. Overview

Key point: A convolution slides a small grid of numbers, the filter, over an image. At each position it multiplies the filter and the pixels under it cell by cell and adds the products. The numbers it writes out form a new grid, the feature map, which is large wherever the image contains the pattern the filter looks for.

A **convolutional neural network** (G-484; CNN) is built from three kinds of layers: convolution layers, pooling layers and fully connected layers (see the [CNN intuition Note](#)). The **convolution layer** (G-480) is the one that gives the network its name, and the one that finds features such as edges.

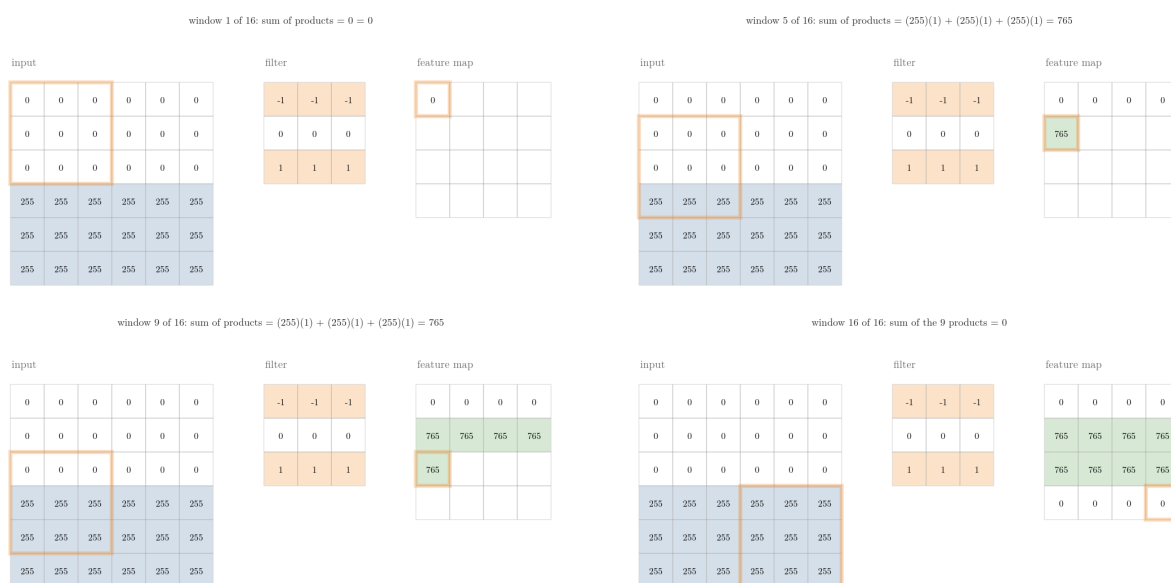


Figure 1: A 3×3 horizontal-edge filter slides over a 6×6 image whose top half is black (0) and bottom half white (255). At each position: multiply cell by cell, add, write one number into the 4×4 feature map. Darker blue means a larger pixel value

Figure 1 shows the whole **convolution operation** (G-481) on a tiny image. This Note covers:

- how images are stored (section 3);
- the sliding window on a list and on a photo: averaging, blur and edges (section 4);
- what an edge is (section 5);
- the convolution step by step (section 6);
- the size of the result (section 7);
- why the filter values are learned (section 8);
- positive and negative values, and ReLU (section 9);
- colour images and many filters (sections 10 and 11).

2. Prerequisites

- The [CNN intuition Note](#): an image is a grid of pixel values, and the first layers of a CNN look for edges.
- The [MNIST ANN Note](#): the MNIST digits and scaling pixels to 0–1.
- The [activation functions Note](#): ReLU.
- The [backpropagation Note](#): how weights are learned.

3. How images are stored

Key point: A greyscale image is one grid of numbers, one per pixel. A colour image is three such grids stacked, one each for red, green and blue: three channels.

3.1 Greyscale images

Key point: One channel. Each pixel holds a number from 0 (black) to 255 (white), often scaled to 0–1.

An **MNIST** (G-1249) digit is a grid of 28×28 **pixels** (G-1501), and each pixel holds one number between 0 and 255 (see the [CNN intuition Note](#) for a digit drawn as its numbers). In MNIST, 0 means black, the background, and 255 means white, the ink. We often divide by 255 to bring the values into the range 0 to 1, as in the [MNIST ANN Note](#).

To the computer, a greyscale image is just a 2D array of numbers. Every operation on the image is an operation on that array.

3.2 Colour (RGB) images

Key point: Three channels, red, green and blue, each a greyscale-like grid of 0–255. Shape: height $\times$ width $\times$ 3.

A colour image stores three grids of the same size, one for each primary colour: **red, green and blue** (RGB). Mixing the three intensities gives every other colour. Each grid is called a **channel** (G-375), and each holds values from 0 to 255.

So a colour image has shape height $\times$ width $\times$ 3. The only difference between a greyscale and an RGB image is the number of channels: one against three.

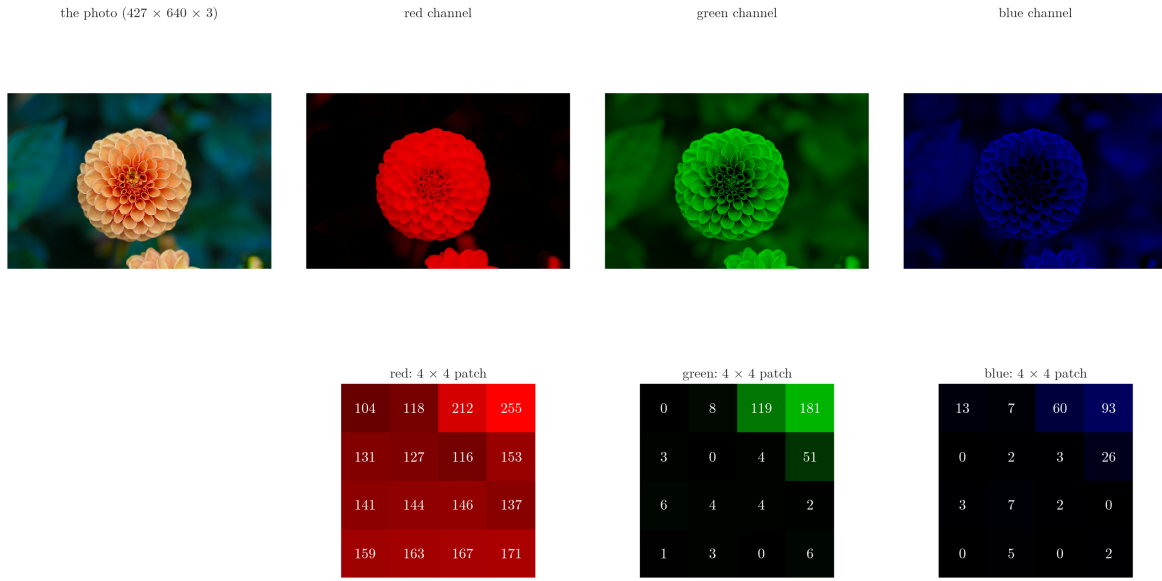


Figure 2: A colour photo (scikit-learn’s sample image flower.jpg) and its three channels. Bottom: the numbers of one 4×4 patch at the flower’s edge, in each channel.

Figure 2 takes a colour photo of 427×640 pixels apart. Each channel is a grid of the same size; drawn on its own, it looks like a greyscale image in one colour. The orange flower is bright in the red channel (values up to 255 in the patch), dimmer in the green one and almost black in the blue one, while the blue-green leaves are the reverse. Mixed back together, the three numbers of each pixel give its colour.

4. A sliding window, first on a list

Key point: Slide a small window of weights along a list of numbers. At each position, multiply each weight by the number under it and add the products. With five weights of $1/5$ the result is a moving average: a smoothed copy of the list. Convolution on an image is the same sliding window in two dimensions.

4.1 A moving average

Key point: A window of five weights of $1/5$ replaces each run of five numbers by their average. Sharp jumps are smoothed out.

Take one row of pixels from the photo of Figure 5: the first 40 pixels of row 100. The values jump up and down as the row crosses bright and dark stripes (Figure 3, grey line).

Now lay a window over the first five pixels, average them, and write the result down. Move the window one pixel to the right and repeat. The list of averages is called a **moving average**.

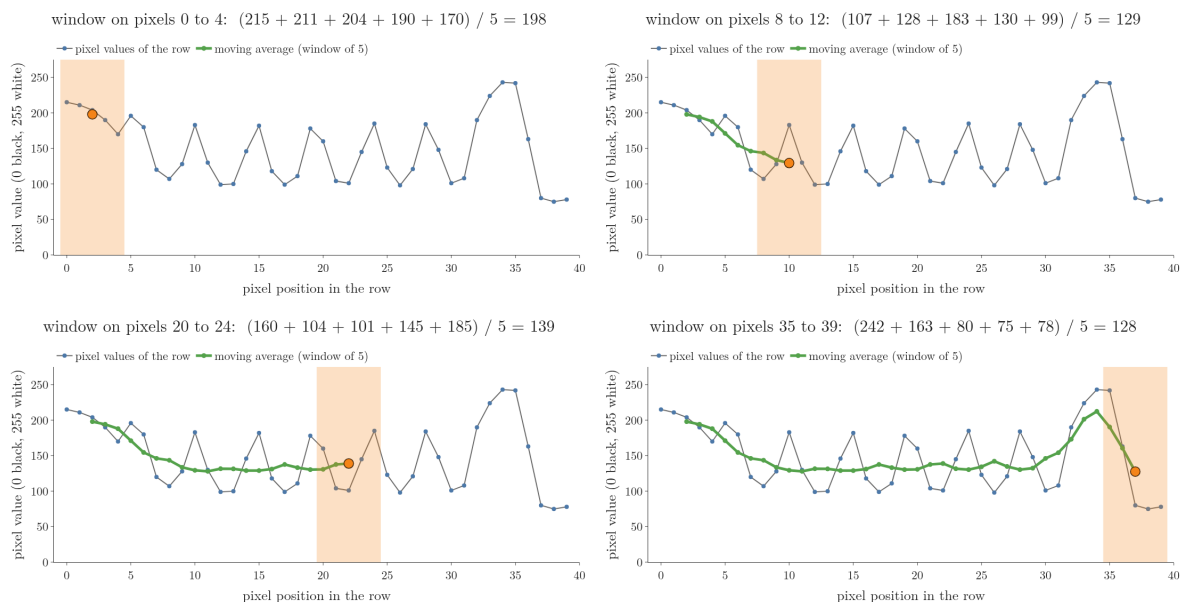


Figure 3: A window of five weights of $1/5$ (orange band) slides along 40 pixels of one row of a photo. At each position it writes the average of the five pixels under it (green). Idea after 3Blue1Brown, “But what is a convolution?”

1. **In words:** multiply each of the five pixels under the window by $1/5$ and add the five products.
2. **Formula:** for pixel values $x_0, x_1, \dots$ and weights $w_0, \dots, w_4$, all equal to $\frac{1}{5}$, the output at position k is

$$z_k = \sum_{m=0}^4 w_m x_{k+m}$$

3. **Example:** the first five pixels are 215, 211, 204, 190 and 170, so $z_0 = (215 + 211 + 204 + 190 + 170)/5 = 198$. One step to the right, the window covers 211, 204, 190, 170 and 196, and $z_1 = 194.2$.

In Figure 3, watch the green line: it follows the grey one but without its sharp teeth. The largest jump between two neighbouring pixels is 83; between two neighbouring averages it is 33.

The small list of weights is the **kernel**, or **filter** (G-777). The operation, slide, multiply, add, is the **convolution operation** (G-481) in one dimension. Nothing else changes when we move to images.

4.2 The same window on an image: blur, then edges

Key point: On an image the kernel is a small grid. If its weights add up to 1, each output pixel is an average of its neighbours and the image blurs. If they add up to 0, flat regions give 0 and only changes in brightness survive: the kernel finds edges.

An image is a grid, so the kernel becomes a small grid too, and it slides both across and down. Figure 4 applies four kernels to the same 64×64 piece of the photo.

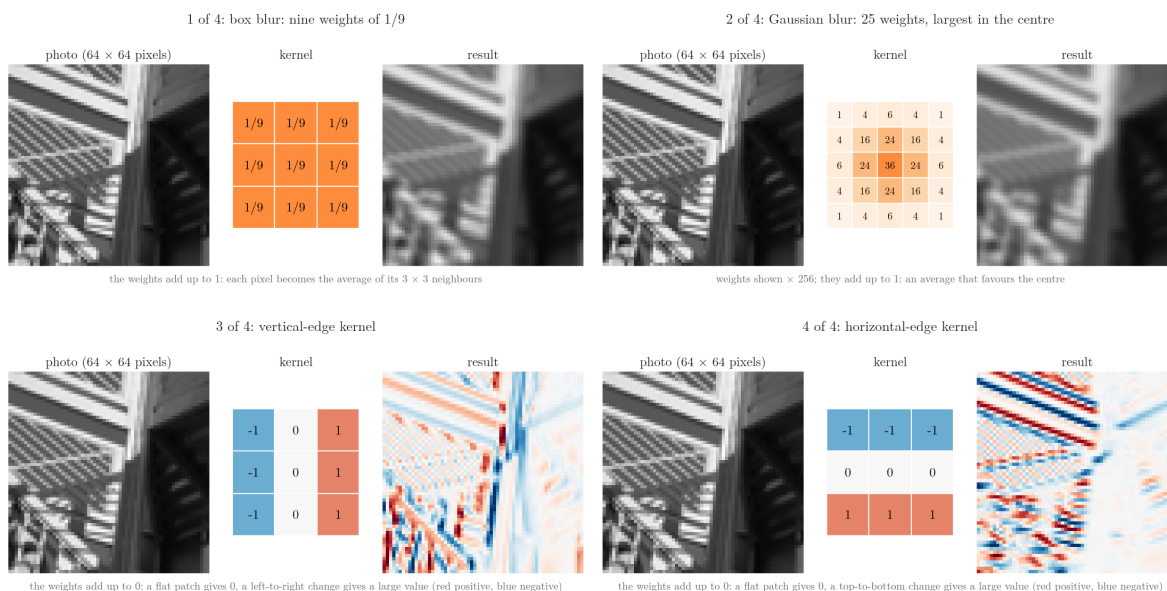


Figure 4: One 64×64 piece of the photo convolved with four kernels, one per frame: box blur, Gaussian blur, vertical-edge kernel, horizontal-edge kernel. Each frame shows the kernel and its result. Sequence of kernels after 3Blue1Brown, “But what is a convolution?”

1. **Box blur:** nine weights of $1/9$. Each pixel becomes the average of its 3×3 neighbourhood, the moving average of section 4.1 in two dimensions. The stripes soften.
2. **Gaussian blur:** a 5×5 kernel whose weights are largest in the centre and small at the rim. Each output pixel is still an average, but one that favours the centre pixel; the blur looks smoother, like a lens out of focus.
3. **Vertical-edge kernel:** -1 in the left column, $+1$ in the right column. The output is the right side of the window minus the left side.
4. **Horizontal-edge kernel:** the same kernel turned by 90 degrees: the bottom of the window minus the top.

What a kernel does can be read from the sum of its weights:

- **Sum 1: an average.** On a flat patch where every pixel is 180, a blur kernel returns 180: the patch is unchanged, and only sharp detail is smoothed.
- **Sum 0: a difference.** On the same flat patch an edge kernel returns $180 \times 0 = 0$. The kernel gives a large value only where the brightness changes across the window.

The photo gives a real example. One nearly flat 3×3 patch of it holds the values 101, 101, 99, 100, 100, 100, 101, 101, 101. The box blur returns their average, 100.4. The vertical-edge kernel returns right column minus left column, $(99 + 100 + 101) - (101 + 100 + 101) = -2$, which is almost 0: no edge here (the script `images/kernel_gallery.py` prints the patch and checks both rules).

Blur is useful in photo editing. A CNN mostly needs the second kind of kernel, the kind that responds to change. Section 5 looks at what such a change is, and section 6 works through an edge kernel by hand.

5. Edges are changes in intensity

Key point: An edge is a place where the brightness changes sharply, from dark to light or light to dark. A filter that compares neighbouring pixels finds it.

Any image is made up of edges: the outline of a face, the side of a building, the stroke of a digit. The first layers of a CNN detect edges, the most basic features of an image (see the [CNN intuition Note](#)).

An **edge** (G-661) is a change in intensity. Where a black region meets a white one, the pixel values jump from 0 to 255. Our eyes see the edge at once; an algorithm must find where the numbers change, and the convolution operation does exactly that.



Figure 5: A real photo (left) and the strength of its edges found by two 3×3 filters (darker = stronger edge). The vertical-edge filter picks out the upright railings; the horizontal-edge filter picks out the horizontal bars and roof edges

Figure 5 shows the result on a real photo. One filter finds the vertical edges, another finds the horizontal ones. Other filters find slanted edges. Section 6 shows how, number by number.

6. The convolution operation

Key point: Place the filter on the top-left corner, multiply each filter value by the pixel under it, add the products, write the sum. Slide one pixel right and repeat; at the end of a row, go back left and one pixel down.

6.1 Filter and feature map

Key point: The filter (or kernel) is a small matrix, usually 3×3 . Convoluting it with the image gives the feature map.

Three objects take part:

- the **image**, the input;
- the **filter** (G-777), also called the **kernel**: a small matrix of numbers, usually 3×3 but sometimes larger;
- the **feature map** (G-766): the output grid of the convolution.

The filter we use is a **horizontal-edge detector** (an **edge detector**, G-659), the fourth kernel of Figure 4:

$$K = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

The filter subtracts the row above from the row below. Its weights add up to 0, so where the two rows are equal it gives 0 (the rule of section 4.2); where the lower row is brighter it gives a large number. The symbol $*$ denotes convolution: $\text{image} * \text{filter} = \text{feature map}$.

6.2 One step

Key point: One position of the filter gives one number: the sum of 9 products.

1. **In words:** lay the filter on a 3×3 window of the image, multiply each pair of overlapping numbers, and add the 9 products.
2. **Formula:** the value at row i , column j of the feature map Z , for an image X and a 3×3 filter K (rows and columns counted from 0):

$$Z_{ij} = \sum_{m=0}^2 \sum_{n=0}^2 X_{i+m, j+n} K_{mn}$$

3. **Example:** the 6×6 image of Figure 1 has three rows of 0 (black) on top of three rows of 255 (white). With the filter on rows 2–4, its top row meets 0s, its middle row meets 0s and its bottom row meets 255s:

$$(-1)(0) \times 3 + (0)(0) \times 3 + (1)(255) \times 3 = 765$$

6.3 Sliding over the whole image

Key point: 16 positions on a 6×6 image give a 4×4 feature map: 0, then two rows of 765, then 0. The large values mark exactly where the edge is.

The filter moves one pixel to the right at a time. At the end of a row it returns to the left and moves one pixel down. Every position gives one number (Figure 1):

- **Row 1** of the feature map (filter on image rows 0–2): all pixels are 0, so every sum is 0.
- **Rows 2 and 3** (filter on rows 1–3 and 2–4): the bottom of the filter sits on white pixels and the top on black or on the boundary, so each sum is $3 \times 255 = 765$.
- **Row 4** (filter on rows 3–5): all pixels are 255. The top row gives -765 , the bottom row $+765$, and they cancel to 0.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 255 & 255 & 255 & 255 & 255 & 255 \\ 255 & 255 & 255 & 255 & 255 & 255 \\ 255 & 255 & 255 & 255 & 255 & 255 \end{bmatrix} * \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 765 & 765 & 765 & 765 \\ 765 & 765 & 765 & 765 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The feature map is 0 in flat regions and large along the band where black meets white: it has found the horizontal edge. Shown as an image, it is a bright stripe across the middle.

Python: The convolution in a few lines of NumPy. `X[i:i+3, j:j+3]` cuts out the 3×3 window; `*` multiplies cell by cell; `.sum()` adds.

```
def conv2d(X, K):
    n_r = X.shape[0] - K.shape[0] + 1
    n_c = X.shape[1] - K.shape[1] + 1
    Z = np.zeros((n_r, n_c))
    for i in range(n_r):
        for j in range(n_c):
            window = X[i:i + K.shape[0], j:j + K.shape[1]]
            Z[i, j] = (window * K).sum()
    return Z
```

The Notebook loads the same filter into a Keras `Conv2D` (G-72) layer: it returns exactly the same 4×4 numbers (largest difference 0).

6.4 Other filters find other edges

Key point: Turn the filter by 90 degrees and it finds vertical edges. Changing the numbers changes the feature the filter looks for.

The **transpose** (G-2012) of the horizontal filter,

$$K_v = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

compares the column on the right with the column on the left, so it is a **vertical-edge detector**. On our 6×6 image, which has no vertical edge, it gives a feature map of all 0s

(Notebook). On the photo of Figure 5 it lights up the railings. With other values we get filters for slanted edges, or for edges with the dark side on the left, right, top or bottom.

Extra: Strictly, mathematicians call this operation **cross-correlation** (G-508). A true convolution first flips the kernel upside down and left to right. On two short lists the flip is easy to see: the true convolution of 1, 2, 3 with 4, 5, 6 turns the second list round to 6, 5, 4, slides it across the first, and gives 4, 13, 28, 27, 18 (NumPy’s `np.convolve`). Most deep learning libraries compute the cross-correlation and call it convolution (Goodfellow et al. 2016, §9.1). The Notebook confirms it for Keras: `Conv2D` gives +765 where the flipped filter would give −765. Because the filter values are learned (section 8), the difference does not matter: training would simply learn the flipped filter.

7. The size of the feature map

Key point: An $n \times n$ image convolved with an $f \times f$ filter gives an $(n - f + 1) \times (n - f + 1)$ feature map.

The filter must fit inside the image. Along one row it can start at positions $0, 1, \dots, n - f$: that is $n - f + 1$ positions. The same holds down the columns.

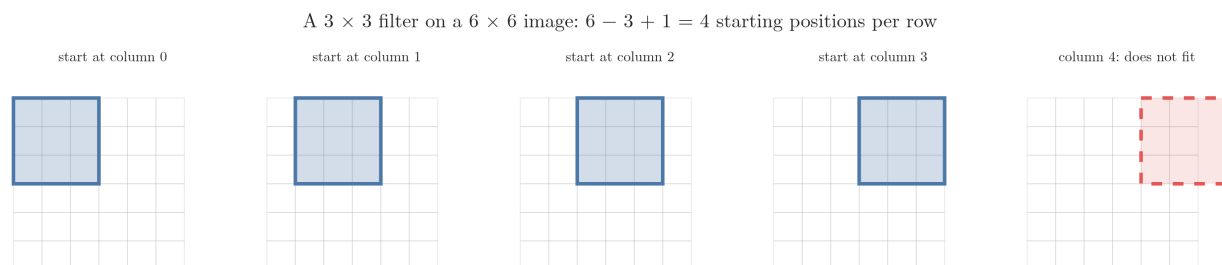


Figure 6: A 3×3 filter on a 6×6 image can start at columns 0 to 3 of a row; at column 4 it would stick out of the image.

Figure 6 shows the four positions along one row. Each position gives one number of the feature map, so each row of the map has 4 numbers.

1. **In words:** the feature map is smaller than the image by the filter size minus 1.
2. **Formula:**

$$\text{output size} = n - f + 1$$

3. **Example:** a 6×6 image and a 3×3 filter: $6 - 3 + 1 = 4$, a 4×4 feature map. An MNIST digit, 28×28 , with a 3×3 filter: $28 - 3 + 1 = 26$. A 64×64 image: 62×62 .

The Notebook checks the formula against Keras for three image sizes and two filter sizes:

Image n	Filter f	$n - f + 1$	Keras Conv2D output
6	3	4	4
6	5	2	2
28	3	26	26
28	5	24	24
64	3	62	62
64	5	60	60

The [padding and strides Note](#) extends the formula to padded images and to filters that jump more than one pixel.

8. Filters are learned, not designed

Key point: In a CNN the filter values are weights. They start random and backpropagation sets them, so the network builds the filters its task needs.

People have designed many filters by hand: left, right, top and bottom edge detectors and others. A CNN does not need them. In deep learning we only choose the filter size and the number of filters; the values start random and are learned during training by **backpropagation** (G-247), exactly like the weights of an ANN (see the [backpropagation Note](#)). The values that come out depend on the training data. Learning its own filters, instead of relying on hand-made ones, is one of the main reasons CNNs became so successful.

The Notebook shows this happening. A small CNN with one convolution layer of 8 filters (3×3), followed by Flatten and a softmax output layer, trains for 2 epochs on MNIST and reaches 96.7% test accuracy.

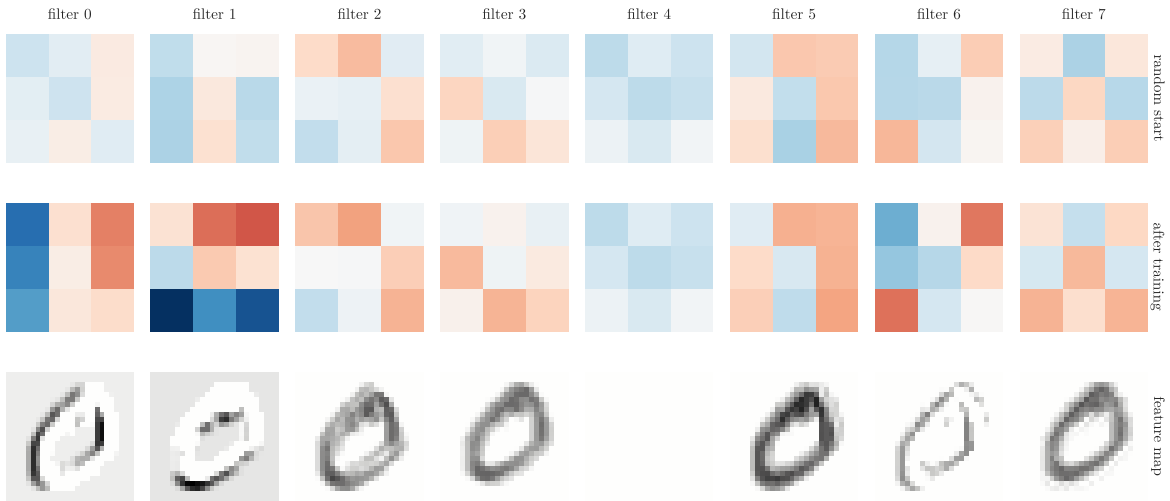


Figure 7: Top: the 8 filters at their random start. Middle: the same filters after training (red positive, blue negative). Bottom: the feature map each trained filter gives for one handwritten 0, after ReLU (darker = larger)

In Figure 7 nobody told the network what to look for, yet filter 0 has turned negative on its left column and positive on its right: a vertical-edge detector, and its feature map marks the left and right sides of the 0. Filter 1 has become positive on top and negative at the bottom, a horizontal-edge detector, and it marks the top and bottom of the 0. Filter 4 ended with all values negative; on this digit, whose pixels are all 0 or more, its map is empty after ReLU.

Extra: Each filter of a `Conv2D` layer also has one bias, added to every value of its feature map before the activation, just as each node of a `Dense` layer has one bias (Keras documentation, `Conv2D`). The [CNN vs ANN Note](#) counts these parameters.

9. Positive and negative values, and ReLU

Key point: A feature map has positive values where the filter’s pattern is present and negative values where its opposite is. ReLU keeps only the positive ones.

Take a **left-edge filter**: positive on its left column, negative on its right, so it gives a large positive value where the image is bright on the left and dark on the right.

$$K_{\text{left}} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

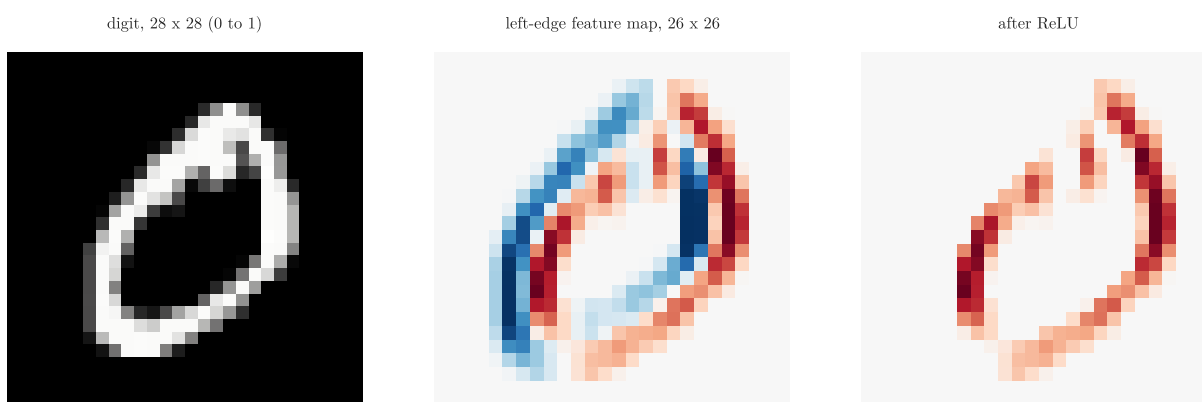


Figure 8: An MNIST 0 (left), its left-edge feature map (middle; red positive, blue negative) and the map after ReLU (right): only the positive, red values remain

On a handwritten 0 (Figure 8, middle), the red values are the places where the filter finds its left edge: the stroke is bright and the background on its right is dark. The blue values are the opposite edge, where the background is on the left: a right edge. The Notebook counts 136 clearly positive cells and 138 clearly negative ones (absolute value above 0.1).

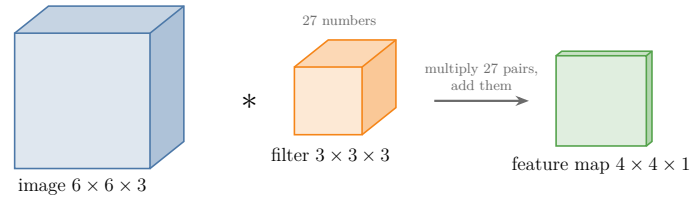
In a CNN, the feature map goes through an activation function next, usually **ReLU** (G-1668), $\max(0, z)$ (see the [activation functions Note](#)). Negative values become 0 and positive values stay. After ReLU (Figure 8, right) only the red left edges remain: the feature map now answers one question, “is there a left edge here?”.

10. Convolution on colour images

Key point: On a 3-channel image the filter has 3 channels too: $3 \times 3 \times 3 = 27$ numbers. Each position still gives one number, so the feature map has 1 channel.

On an RGB image, a 3×3 filter is automatically a $3 \times 3 \times 3$ filter: one 3×3 slice for each of the red, green and blue channels. Think of the image as a box of depth 3 and the filter as a small box of the same depth.

One filter: a 3-channel image gives one feature map



Two filters: two feature maps, stacked

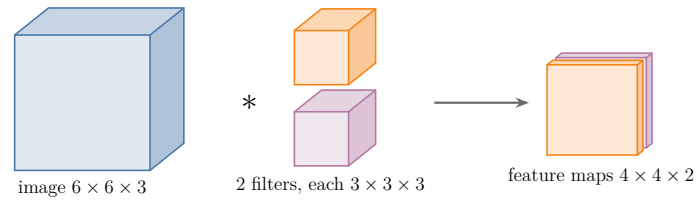


Figure 9: Top: a $6 \times 6 \times 3$ image convolved with one $3 \times 3 \times 3$ filter gives one 4×4 feature map. Bottom: two filters give two feature maps, stacked into a $4 \times 4 \times 2$ volume

The small box slides over the large one exactly as before (Figure 9, top). At each position it multiplies 27 pairs of numbers, 9 in each channel, and adds all 27 products into a single number. So the output has one channel.

1. **In words:** the filter covers all channels at once and sums over them, so one filter gives one single-channel feature map.
2. **Formula:** an $n \times n \times c$ image and an $f \times f \times c$ filter give

$$(n - f + 1) \times (n - f + 1) \times 1$$

3. **Example:** a $6 \times 6 \times 3$ image and a $3 \times 3 \times 3$ filter give a $4 \times 4 \times 1$ feature map. The Notebook checks that Keras' result equals convolving each channel separately with its slice of the filter and adding the three maps.

The number of channels of the filter always equals the number of channels of its input; we never choose it.

11. Many filters

Key point: Each filter gives one feature map. With k filters, the maps stack into a volume of depth k , which is the input of the next layer.

A convolution layer almost never uses a single filter. It may use one filter for vertical edges, one for horizontal edges, one for slanted edges, and so on. Each filter is convolved with the same image and gives its own feature map of the same size.

The maps are stacked into a volume (Figure 9, bottom):

Input	Filters	Output
$6 \times 6 \times 3$	1 of $3 \times 3 \times 3$	$4 \times 4 \times 1$
$6 \times 6 \times 3$	2 of $3 \times 3 \times 3$	$4 \times 4 \times 2$
$6 \times 6 \times 3$	10 of $3 \times 3 \times 3$	$4 \times 4 \times 10$

The number of filters becomes the number of channels of the output. The Notebook gets these three shapes from Keras. The output volume works like an image with k channels for the next convolution layer, whose filters then have depth k .

Python: In Keras the number of filters and the filter size are the first two arguments of Conv2D; the depth of each filter follows from the input.

```
layer = keras.layers.Conv2D(10, 3)    # 10 filters, each 3 x 3
layer(img[None]).shape                 # img: 6 x 6 x 3
# (1, 4, 4, 10): batch, height, width, channels
```

12. Summary

Object	Shape	Meaning
Greyscale image	$n \times n \times 1$	one number per pixel, 0 (black) to 255 (white)
RGB image	$n \times n \times 3$	red, green and blue channels
Filter (kernel)	$f \times f \times c$	learned weights; depth $c =$ input channels
Feature map	$(n - f + 1) \times (n - f + 1)$	one per filter
Output of a layer with k filters	$(n - f + 1) \times (n - f + 1) \times k$	the maps stacked

- Convolution: slide the filter, multiply cell by cell, add, write one number per position.
- A kernel whose weights add up to 1 averages (blur); one whose weights add up to 0 responds only to change (edges).
- An edge is a change in intensity; edge filters give large values where the image changes and 0 where it is flat.
- Output size $n - f + 1$; on a colour image one filter still gives one feature map.
- Filter values are learned by backpropagation, like ANN weights; a small CNN on MNIST learned vertical- and horizontal-edge detectors on its own.
- ReLU after the convolution keeps the positive responses only.

13. Sources

Built from

- CampusX, “CNN Part 3 | Convolution Operation”, YouTube, <https://www.youtube.com/watch?v=cgJ...>
- Sanderson, G. (3Blue1Brown), “But what is a convolution?”, 2022, <https://www.youtube.com/watch?v...> (the moving average, blur kernels and the sum-to-1 and sum-to-0 rule of section 4; the flip in true convolution)

Other references

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Chapter 9, Convolutional Networks, §9.1 (the convolution operation, cross-correlation). deeplearningbook.org.
- Keras API documentation: **Conv2D** layer, keras.io/api/layers/convolution_layers/convolution2d.
- Stanford CS231n, Convolutional Neural Networks for Visual Recognition, course notes “Convolutional Neural Networks: Architectures, Convolution / Pooling Layers”, cs231n.github.io/convolutional-networks.
- scikit-learn developers. `load_sample_image("flower.jpg")`, one of scikit-learn’s two sample photos, photo by danielbuechele (Flickr), CC BY 2.0, as listed in `load_sample_images().DESCR`.
- SciPy **ascent** test image (public domain), from the `scipy/dataset-ascent` repository, used for Figures 3, 4 and 5.

14. Key terms

Term	Meaning
Pixel	One cell of an image grid, holding an intensity value
Channel	One grid of an image: 1 for greyscale, 3 (red, green, blue) for colour
Edge	A place where the intensity changes sharply
Filter (kernel)	A small matrix of weights slid over the image; its depth equals the input's channels
Convolution operation	Sliding a filter over an input, multiplying cell by cell and adding at each position
Feature map	The grid of numbers a filter produces; large where the filter's pattern is present
Edge detector	A filter whose feature map is large along edges of one direction
Moving average	The list of averages a sliding window gives: each output is the mean of the numbers under the window
Cross-correlation	Convolution without flipping the kernel; what deep learning libraries compute
Conv2D	The Keras layer for 2D convolution: <code>Conv2D(filters, kernel_size)</code>